Anjali S Rameja, PhD

Computational Materials Scientist

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SIMULATION

- Molecular dynamics, atomistic and particle-based simulation (LAMMPS)
- Density Functional Theory (Quantum ESPRESSO)

Familiar (in progress): ML interatomic potentials (MACE), machine learning for materials, Monte Carlo

INTERATOMIC POTENTIALS AND MATERIALS DOMAIN

- 2nd nearest neighbor MEAM
- Embedded Atom Method (EAM)
- Stillinger-Weber
- Tight binding
- Ultra Soft Pseudo Potential
- Liquid and Glassy Metallic Alloys
- Liquid Transition Metals

PROGRAMMING AND COMPUTING

- Python (NumPy, Pandas, SciPy)
- Git and version control
- Shell Scripting
- HPC and Linux

MODELING AND ANALYSIS

- Statistical mechanics and mathematical modeling
- Liquid-state theory
- Green-Kubo and Stokes-Einstein transport (diffusion, viscosity)
- Structure and phase transitions (glass transition, RDF, CNA)
- Time correlation functions
- Thermophysical and mechanical properties

TOOLS

- Visualization tools (OVITO, VESTA, XCrySDen)
- Data Analysis (MS-Excel, QtiPlot, Python)
- Generative AI (Claude, Gemini, ChatGPT)

PROFESSIONAL SUMMARY

Computational scientist with a Ph.D. in physics and 6+ years in atomistic and molecular simulation of disordered, liquid, and glassy systems. Expert in classical molecular dynamics and molecular modeling (LAMMPS, 10,000-atom systems, 300 to 5800 K), with hands-on DFT (Quantum ESPRESSO) and Python-driven analysis on Linux HPC. Six peer-reviewed publications including a Featured Article in Journal of Applied Physics. Deep background in transport, thermophysical, and structural properties, glass transition, and supercooled-liquid dynamics, directly transferable to molecular modeling and materials R&D. Currently evaluating frontier generative AI on materials science problems as a Research Quality Specialist at Turing.

CURRENT PROJECT

- Machine-learning interatomic potential (MACE): fine-tuning MACE-MP-0 on DFT data for Mg-Ca, benchmarked against my 2NN-MEAM results. In progress.

PROFESSIONAL EXPERIENCE

Research Quality Specialist, Materials Science

Turing | Remote | June 2026 to Present

- Evaluate generative AI on research level materials science and physics problems.
- Benchmark physical reasoning, scientific rigor, and technical accuracy of AI generated content.

Doctoral Researcher, Computational Materials Science

Department of Physics, Maharaja Krishnakumarsinhji Bhavnagar University | Feb 2020 to Jan 2026

Supervisor: Prof. N. K. Bhatt

Transport and Thermophysical Properties of Mg-Ca and Al-Ca Bulk Metallic Glasses for Biomedical Applications

- Designed and executed large-scale LAMMPS molecular-dynamics simulations on Mg_2Ca, Al_2Ca, Al_4Ca, and $Ca_{61}Al_{39}$ systems (10,000 atoms, 300 to 1500 K) using 2NN-MEAM interatomic potentials.
- Quantified diffusion coefficients, viscosity, glass-transition temperatures, elastic moduli, thermodynamic properties, and structural descriptors (RDF, CNA) across glassy, supercooled, and liquid regimes.
- Demonstrated breakdown of the Stokes-Einstein relation in supercooled BMGs.
- Established Mg-Ca BMGs as candidate biodegradable bone-grafting implants via comparative analysis with experimental biomaterials data.

Structural and Transport Properties of Liquid Transition Metals (W, Mo)

- Conducted MD simulations of liquid tungsten (3700 to 5800 K) and molybdenum (2900 to 4900 K) using tight-binding and Stillinger-Weber potentials.
- Characterized self-diffusion, viscosity, and structural ordering across the full liquid temperature range from melting to boiling point.

EDUCATION

PhD, Physics (Computational Materials Science)
MK Bhavnagar University, Gujarat
2020 to 2026

MSc, Physics (Condensed Matter)
Sardar Patel University, Gujarat
2013 to 2015

BSc, Physics (ranked 3rd in cohort)
Sir P. P. Institute of Science, Gujarat
2010 to 2013

ACHIEVEMENTS

1st Prize: Oral Presentation
National Conference at IITE, Gandhinagar, Gujarat | 2022

2nd Prize: Poster Presentation
National Science Symposium, Govt. Sci. College, Gariyadhar, Gujarat | 2024

3rd Rank: BSc Physics Cohort

First-Principles DFT Study of Mg-Al-Ca System

- Quantum ESPRESSO calculations on Mg_2Ca, Al_2Ca, Al_4Ca crystalline phases for electronic structure, band structure, and chemical-bonding analysis.
- Cross-validated MD-derived thermophysical properties against ab-initio reference.

Python Automation for Simulation Analysis

- Built reusable Python pipelines (NumPy, SciPy, Pandas) for batch post-processing of multi-temperature MD trajectories: RDF, MSD, VACF, structure factor, and time-correlation function analysis.
- Cut post-processing turnaround per simulation set compared to manual workflows.

PUBLICATIONS

1. A. Shankar, N. K. Bhatt. "Atomic mass transport in binary metallic glasses: a case study of Mg_2Ca, Al_2Ca and Al_4Ca." Journal of Physics: Condensed Matter 37, 355401 (2025).
2. A. Shankar, N. K. Bhatt. "Enhanced mechanical and thermophysical properties of Mg_2Ca, Al_2Ca, and Al_4Ca bulk metallic glasses in comparison to crystalline alloys for bone grafting applications." Journal of Applied Physics 136(3) (2024). [Featured Article]
3. A. Shankar, N. K. Bhatt. "Bulk metallic glass Al_2Ca from a biomedical application viewpoint: A molecular dynamics study." IOP Conference Series: Materials Science and Engineering 1291, 012026 (2023).
4. D. R. Gohil, A. Shankar, N. K. Bhatt. "Mass transport and thermal properties of liquid (melting to boiling point) tungsten: a molecular dynamics simulation." Physica Scripta 98, 115963 (2023).
5. D. R. Gohil, A. Shankar, N. K. Bhatt. "Molecular Dynamics Simulations for Liquid Molybdenum: Structural and Atomic Transport Properties." Bulgarian Journal of Physics 50(4) (2023).
6. A. Shankar, N. K. Bhatt, A. B. Patel, P. R. Vyas. "Amorphous structure of binary $Ca_{61}Al_{39}$ metallic glass close to transition temperature." AIP Conference Proceedings 2352, 020038 (2021).

HIGHLIGHTS

- 16th International Conference on Creep and Fracture of Engineering Materials and Structures | IISc | 2024
- Young Researchers' Workshop on Machine Learning for Materials | ICTP | 2022
- Molecular Dynamics Simulations in Materials Science and Engineering | IITkgp | 2021
- Computational Physics and Materials Science: Total Energy and Force Methods | ICTP | 2021
- Introduction to Glass Science and Technology | IITkgp | 2020
- 9 conference presentations (2020 to 2024).

OTHER EXPERIENCE

Visiting Faculty and Lecturer, Physics

MK Bhavnagar University and affiliated colleges | 2015 to 2024

- Taught computational and condensed matter physics (MSc), supervised LAMMPS-based MD dissertations, ran MD and visualization workshops, led departmental initiatives.